

# Determinants

The goal is to come up with a way to talk about a notion of generalized volume in  $n$ -space. It will have the property that if the volume determined by the  $n$  columns of a matrix is not zero then that matrix will be invertible, and if the volume is zero then it will not be invertible.

The  $2 \times 2$  case:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

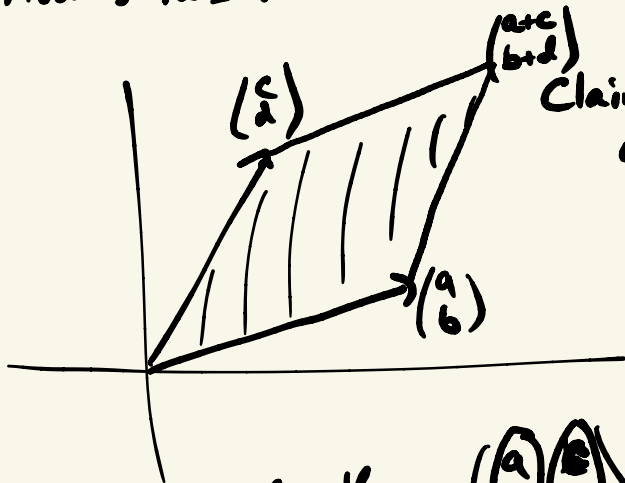
*A is invertible*

*iff  $ad - bc \neq 0$*

When is  $A$  invertible? Assume  $a \neq 0$ .

*If  $ad - bc \neq 0$  then  $A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$*

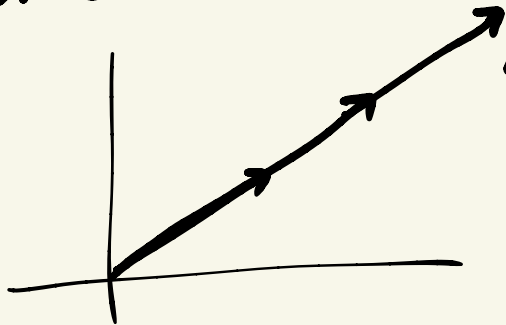
How is this related to area?



Claim: The area of the parallelogram shown is  $|ad - bc|$ .

If  $ad - bc = 0$  then  $\begin{pmatrix} a & c \\ b & d \end{pmatrix}$  is not invertible.

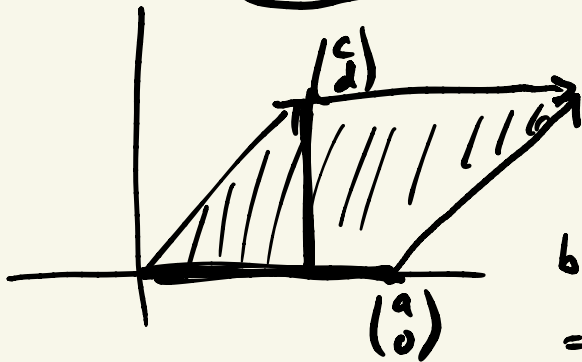
So  $\begin{pmatrix} a \\ b \end{pmatrix}$  and  $\begin{pmatrix} c \\ d \end{pmatrix}$  are scalar multiples of each other.



← the area of  
this "parallelogram"  
is 0.

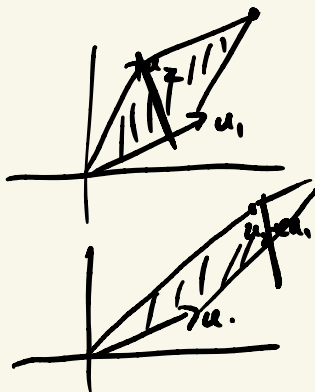
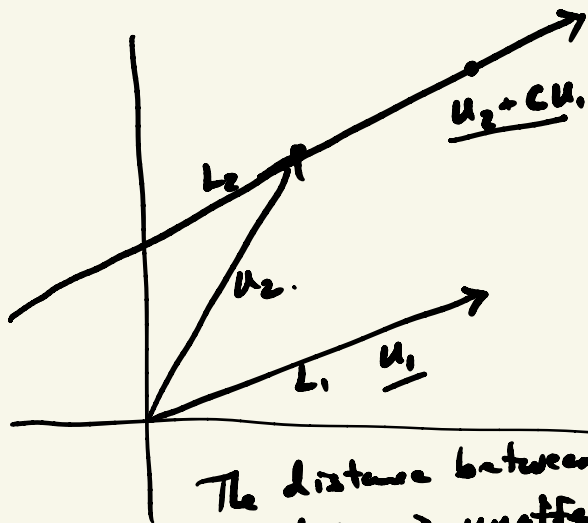
Another case:

$$\begin{pmatrix} a & c \\ \underline{0} & d \end{pmatrix} \leftarrow \text{looks good.}$$



What's the area?

$$\text{base} \times \text{height} \\ = |ad|$$



The distance between  $L_1$  and  $L_2$  is unaffected by  $c$ .

By choosing  $c$  appropriately we can change the matrix to one in the form

$$\begin{pmatrix} a & c \\ 0 & d \end{pmatrix} \text{ or } \begin{pmatrix} a & 0 \\ b & d \end{pmatrix}$$

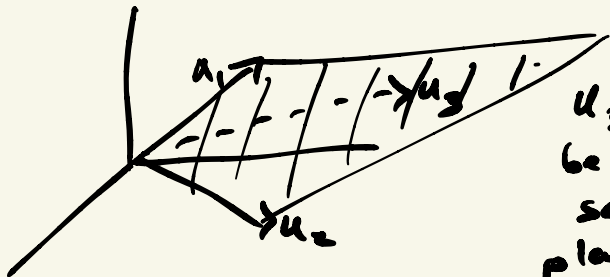
Conclusion: The area of the parallelogram is  $|ad - bc|$ .

Notation:  $\det \begin{pmatrix} a & c \\ b & d \end{pmatrix} = ad - bc$ .

What happens in  $\mathbb{R}^3$ ?

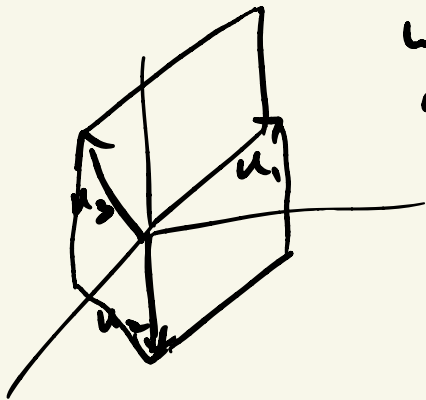
Suppose  $A = \begin{pmatrix} u_1 & u_2 & u_3 \end{pmatrix} \xleftarrow{3 \times 3} \underline{\underline{\text{col}}}$

$A$  is not invertible.

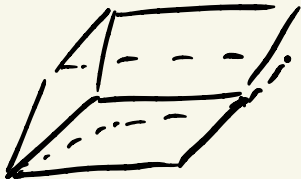


$u_3$  might  
be in the  
same  
plane as  
 $u_1$  and  $u_2$ .

In general we would have.



We want the parallelepiped  
determined by  $u_1, u_2$   
and  $u_3$ .



Claim is that the number, in absolute value, we  
want is the volume of this solid.

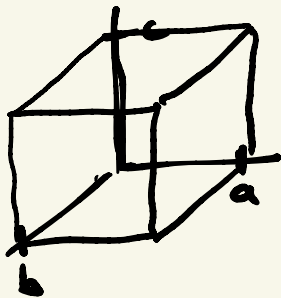


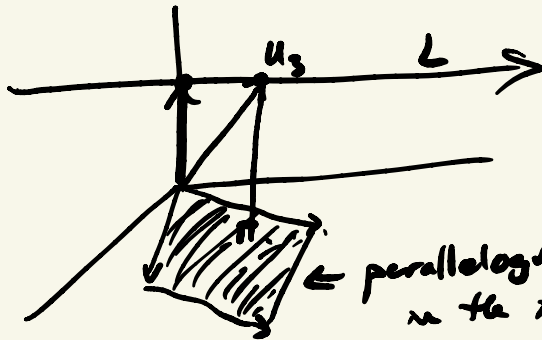
Notice if  $A$  is not invertible then the volume of the parallelepiped is 0.

Ex: The simplest case

$$A = \begin{pmatrix} a & b & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{pmatrix}$$

Volume here is  $|abc|$ .





$L$  is parallel  
to the  $xy$ -plane  
then using  
 $u_3$  of  
any point  
on  $L$  will

← parallelogram  
in the  $xy$ -plane.

give the same  
volume.

We will determine volume in  $\mathbb{R}^{n+1}$  by induction on knowing what volume means in  $\mathbb{R}^n$  as we did to go from  $\mathbb{R}^2$  to  $\mathbb{R}^3$ .

# The determinant as area

The area of the parallelogram formed by  $\begin{bmatrix} a \\ c \end{bmatrix}$  and  $\begin{bmatrix} b \\ d \end{bmatrix}$  is  $|ad - bc|$ , the absolute value of the determinant of the matrix  $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ .

# The 3 x 3 case and volume

The absolute value of the determinant of a 3 x 3 matrix is the volume of the parallelepiped formed by the three columns in  $\mathbb{R}^3$ .

## Definition

- 1 Suppose  $A = (a_{ij})$  is an  $n \times n$  matrix. Then the submatrix  $A_{ij}$  is obtained by removing the  $i^{th}$  row and  $j^{th}$  column from  $A$ . The  $(i, j)$ -cofactor is

$$C_{ij} := (-1)^{i+j} \det(A_{ij}).$$

- 2 We define the determinant inductively: if  $A$  is an  $n \times n$  matrix and  $1 \leq i \leq n$  then

$$\det(A) = a_{i1}C_{i1} + a_{i2}C_{i2} + \dots + a_{in}C_{in}.$$

Equivalently, for any  $1 \leq j \leq n$ ,

$$\det(A) = a_{1j}C_{1j} + a_{2j}C_{2j} + \dots + a_{nj}C_{nj}.$$

$$\text{Ex: } A = \begin{pmatrix} 1 & -1 & 2 \\ 3 & 1 & 4 \\ 2 & 0 & 1 \end{pmatrix}$$

$$A_{11} = \begin{pmatrix} 1 & 4 \\ 0 & 1 \end{pmatrix}$$

$$C_{11} = \underline{(-1)^{1+1}} \det(A_{11})$$
$$= 1$$

$$A_{32} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$

$$C_{32} = (-1)^{3+2} \det(A_{32})$$
$$= -(-2) = 2.$$